

Case Study: UI Development for Smart EV Charging Station Interface

Industry: Electric Vehicles / Clean Energy

Service Delivered: Frontend Development for Charging Station Platform

Overview

An electric mobility company collaborated with **Softcode Technologies** to design and develop a modern user interface (UI) for their EV charging station system. The primary goal was to enable **authenticated EV users** to quickly and easily:

- 📖 **Locate nearby charging stations**
- 🔄 **View real-time availability**
- ⌚ **Know whether a station is free at a specific time**

The interface needed to be **intuitive, responsive**, and designed for **touch-based interaction**, compatible with both web and tablet/kiosk screens.

Objectives

The frontend solution needed to offer:

- 🗝️ **User Authentication:** Only authorized users can access charging station information
- 📍 **Station Locator:** A live map showing stations near the user's location
- ✅ **Availability Status:** Instant visual indicators of which stations are free, occupied, or reserved
- 📅 **Time-Based Availability:** Check if the station will be available at a future time

Key Features Implemented



Clean, Minimal UI

- Developed using **React.js** and **Tailwind CSS** for rapid performance and scalable design
- Focused on **user-centric flows**, enabling easy navigation and interaction even while driving



Live Data Integration

- Pulled real-time data from backend APIs to reflect **station status (free/busy)**
- Used **color-coded indicators** and live updates to help users plan charging stops efficiently

Distance & ETA Display

- Integrated with **Google Maps API** to calculate how far each station is from the user
- Displayed **estimated time to reach**, improving travel decisions

Login & Access Control

- Developed a secure **login interface for authenticated users**
- Role-based UI elements (admin vs. EV user) for future extensibility

Results

Improved User Experience

EV owners found it much easier to plan charging sessions, reducing downtime and frustration.

Higher Station Utilization

The transparent, real-time availability data helped users avoid congested or unavailable stations.

Scalable UI Architecture

The UI was built in a way that supports easy future expansion — like scheduling, payment integration, or admin dashboards.

Conclusion

Softcode's design-first and user-focused frontend solution helped create a smoother, smarter experience for EV users. By combining real-time station data with intuitive design, we enabled users to make quick and confident decisions while on the move.

Looking to build a UI for your EV or IoT platform?

Let's bring your idea to life — krishnasharma@softcode.io